

2.1 Cart-Pole Swing-Up with Limited Actuation

2.1(a)

We want the constraint $s_{k+1} = f_d(s_k, u_k)$ to have the form $s_{k+1} = A_k s_k + B_k u_k + c_k$ to derive the convex approximation of the OCP at iteration i , around $(s_k^{(i)}, u_k^{(i)})$.

The linearization of $f_d(s_k, u_k)$ becomes:

$$f_d(s_k^{(i)}, u_k^{(i)}) = f(s_k^{(i)}, u_k^{(i)}) + \frac{\partial f_d}{\partial s}(s_k^{(i)}, u_k^{(i)})(s_k - s_k^{(i)}) + \frac{\partial f_d}{\partial u}(s_k^{(i)}, u_k^{(i)})(u_k - u_k^{(i)})$$

After expanding terms, we come up with

$$A_k = \frac{\partial f_d}{\partial s}(s_k^{(i)}, u_k^{(i)}), \quad B_k = \frac{\partial f_d}{\partial u}(s_k^{(i)}, u_k^{(i)}),$$

$$c_k = f_d(s_k^{(i)}, u_k^{(i)}) - A_k s_k^{(i)} - B_k u_k^{(i)}.$$

The convex subproblem is

$$\begin{aligned} \min_{s, u} \quad & \sum_{k=0}^{N-1} ((s_k - s_{\text{goal}})^T Q (s_k - s_{\text{goal}}) + u_k^T R u_k) + (s_N - s_{\text{goal}})^T P (s_N - s_{\text{goal}}) \\ \text{s.t.} \quad & s_0 = \bar{s}_0, \\ & s_{k+1} = A_k s_k + B_k u_k + c_k, \\ & -u_{\max} \leq u_k \leq u_{\max}, \\ & \|s_k - s_k^{(i)}\|_{\infty} \leq \rho, \quad \|u_k - u_k^{(i)}\|_{\infty} \leq \rho. \end{aligned}$$

2.1(b)

```

1 A, B = jax.jacobian(f, argnums=(0, 1))(s, u)
2 c = f(s, u) - A @ s - B @ u

```

2.1(d)

First use SCP to compute a feasible swing-up trajectory. Once the pendulum is near $(0, \pi, 0, 0)$, switch to a local LQR controller designed around $(0, \pi, 0, 0)$ for stabilization.

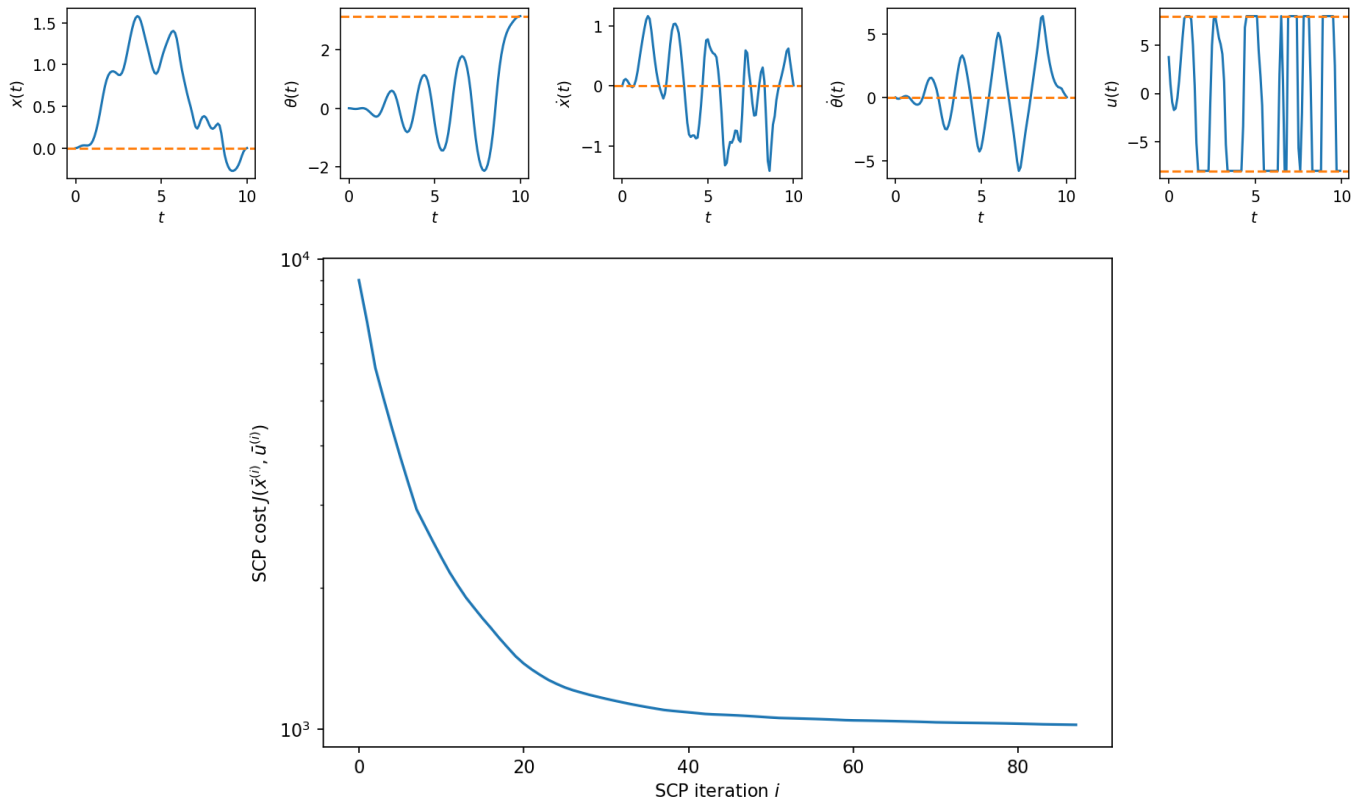


Figure 1: SCP swing-up trajectory with bounded input and trust-region updates.

Code

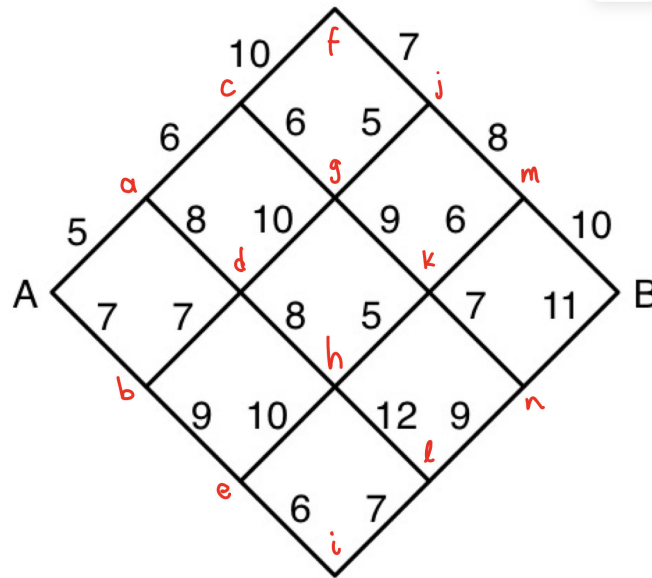
Code in Appendix: `cartpole_swingup_constrained.py`.

2.2 Shortest Path Through a Grid

2.2(a)

Following DP, the next table, based on Fig. , with each node cost is computed.

node	$J(\text{node})$
B	0
m	10
n	11
j	18
k	16
l	20
f	25
g	23
h	21
i	27
c	29
d	29
e	31
a	35
b	36
A	40



The shortest path has cost $\boxed{40}$. And the optimal path is:

$$A \rightarrow a \rightarrow c \rightarrow g \rightarrow j \rightarrow m \rightarrow B.$$

2.2(b)

For an n -segment grid, each route consists of n up moves and n down moves, so exhaustive search evaluates

$$\boxed{\binom{2n}{n}} \text{ routes.}$$

Dynamic programming evaluates each nonterminal node. The number of DP evaluations is $n(n+2)$.

For $n = 3$, these give $\binom{6}{3} = 20$ and $3(5) = 15$.

2.3 Cart-Pole Balance

2.3(a)

With $\tilde{s}_k = s_k - \bar{s}$, Euler discretization gives

$$\tilde{s}_{k+1} \approx A\tilde{s}_k + Bu_k$$

where

$$A = I + \Delta t \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{m_p g}{m_c} & 0 & 0 \\ 0 & \frac{(m_c + m_p)g}{m_c \ell} & 0 & 0 \end{bmatrix}, \quad B = \Delta t \begin{bmatrix} 0 \\ 0 \\ \frac{1}{m_c} \\ \frac{1}{m_c \ell} \end{bmatrix}$$

2.3(b)

Using Riccati recursion until $\|P_k - P_{k-1}\|_{\max} < 10^{-4}$ gives

$$K: \begin{bmatrix} 0.73 & -231.85 & 4.22 & -68.24 \end{bmatrix}$$

2.3(c)

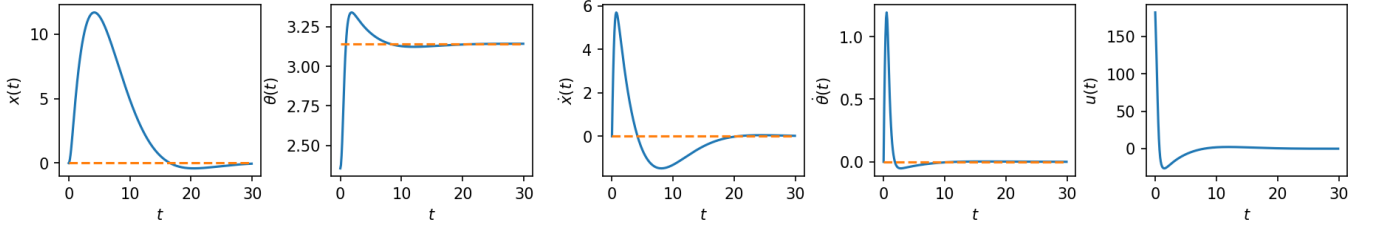


Figure 2: Infinite-horizon LQR controller for cart-pole balance

2.3(d)

i. We can reuse A and B because the dynamics Jacobians along the reference do not depend on x , and the desired trajectory keeps $\theta = \pi$, $\dot{\theta} = 0$. Then the same upright linearization is valid along the reference.

ii.

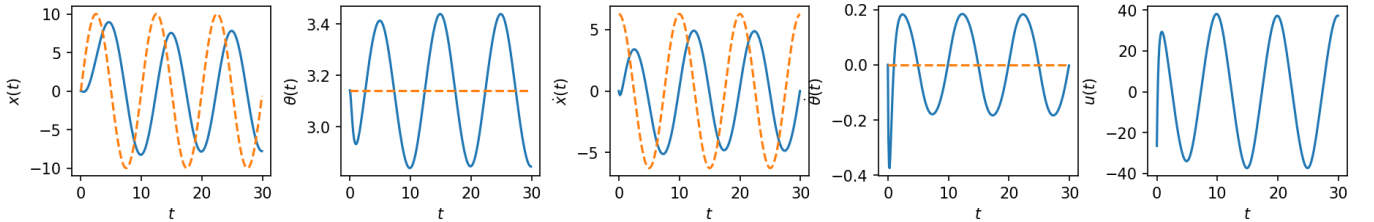


Figure 3: Tracking new reference trajectory $s(t)$ with LQR feedback.

iii. The pendulum must tilt to generate the horizontal cart acceleration from the reference, so keeping $\theta(t) \equiv \pi$ while moving the cart sinusoidally is not a dynamically consistent upright trajectory without feedforward compensation.

Code

Code in Appendix: `cartpole_balance.py`.

2.4 Cart-Pole Swing-Up

2.4(a)

The requested one-line JAX linearization is

```
1 A, B = jax.jacobian(f, args=(0, 1))(s, u)
```

2.4(b)

Let

$$\tilde{s}_k = s_k - \bar{s}_k, \quad \tilde{u}_k = u_k - \bar{u}_k.$$

Expanding the quadratic cost gives linear terms

$$q_N = Q_N(\bar{s}_N - s_{\text{goal}}), \quad q_k = Q(\bar{s}_k - s_{\text{goal}}), \quad r_k = R\bar{u}_k.$$

2.4(c)

The iLQR implementation is in `code/cartpole_swingup.py`. It computes linearizations, performs the Riccati backward pass for feedback gains Y_k and offsets y_k , updates the nominal trajectory, and uses backtracking to keep the update stable.

2.4(d)

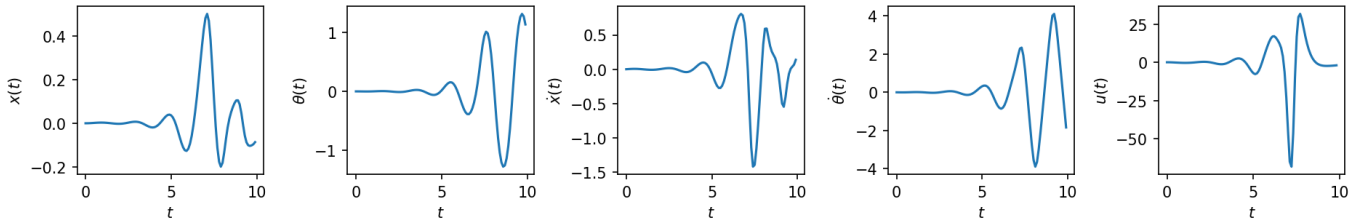


Figure 4: Open-loop application of the iLQR control sequence.

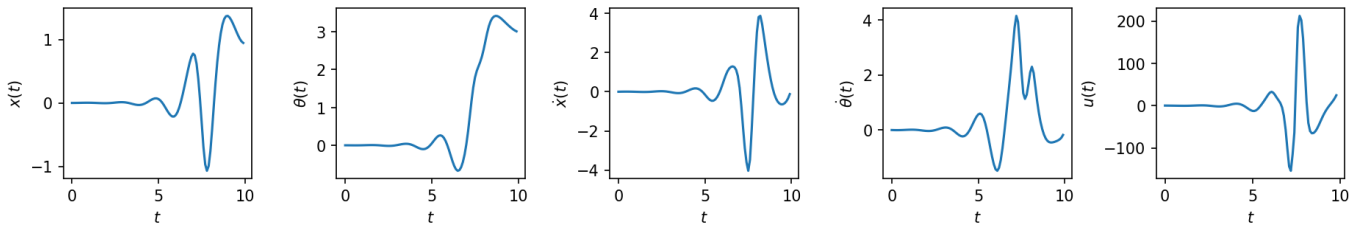


Figure 5: Closed-loop iLQR policy.

2.5 Machine Maintenance

Let $V_R(h)$ and $V_B(h)$ denote the optimal expected profit with h weeks remaining when the machine starts the week running or broken.

$$\begin{aligned} V_R(h) &= \max\{40 + 0.6V_R(h-1) + 0.4V_B(h-1), \\ &\quad 30 + 0.3V_R(h-1) + 0.7V_B(h-1)\}, \\ V_B(h) &= \max\{20 + 0.6V_R(h-1) + 0.4V_B(h-1), -50 + V_R(h-1)\}. \end{aligned}$$

With $V_R(0) = V_B(0) = 0$, the computed values are

h	0	1	2	3
$V_R(h)$	0	40	72	104
$V_B(h)$	0	20	52	84

The optimal policy is to perform preventive maintenance when running and to repair when broken. Since the machine is new at week 1 and guaranteed to run through that week, the total expected profit over four weeks is

$$\boxed{100 + V_R(3) = 204}.$$

Code Appendix

Homework part	Code file
2.1(b), 2.1(c), 2.1(d)	cartpole_swingup_constrained.py
2.3(a), 2.3(b), 2.3(c), 2.3(d)	cartpole_balance.py
2.4(a), 2.4(c), 2.4(d)	cartpole_swingup.py

cartpole_swingup_constrained.py

Answers 2.1(b), 2.1(c), and 2.1(d).

```
1 """
2 Starter code for the problem "Cart-pole swing-up with limited actuation".
3
4 Autonomous Systems Lab (ASL), Stanford University
5 """
6
7 from functools import partial
8 from pathlib import Path
9
10 from animations import animate_cartpole
11
12 import cvxpy as cvx
13
14 import jax
15 import jax.numpy as jnp
16
17 import matplotlib.pyplot as plt
18
19 import numpy as np
20
21 from tqdm import tqdm
22
23 FIG_DIR = Path(__file__).resolve().parents[1] / "latex" / "figures"
24 FIG_DIR.mkdir(parents=True, exist_ok=True)
25
26 @partial(jax.jit, static_argnums=(0,))
27 @partial(jax.vmap, in_axes=(None, 0, 0))
28 def affinize(f, s, u):
29     """Affinize the function f(s, u) around (s, u).
30
31     Arguments
32     -----
33     f: callable
34         A nonlinear function with call signature f(s, u).
35     s: numpy.ndarray
36         The state (1-D).
37     u: numpy.ndarray
38         The control input (1-D).
39
40     Returns
41     -----
42     A: jax.numpy.ndarray
43         The Jacobian of f at (s, u), with respect to s.
44     B: jax.numpy.ndarray
45         The Jacobian of f at (s, u), with respect to u.
46     c: jax.numpy.ndarray
47         The offset term in the first-order Taylor expansion of f at (s, u)
48         that sums all vector terms strictly dependent on the nominal point
49         (s, u) alone.
```

```

50 """
51 # PART (b) #####
52 # INSTRUCTIONS: Use JAX to affimize 'f' around '(s, u)' in two lines.
53 A, B = jax.jacobian(f, argnums=(0, 1))(s, u)
54 c = f(s, u) - A @ s - B @ u
55 # END PART (b) #####
56 return A, B, c
57
58
59 def solve_swingup_scp(f, s0, s_goal, N, P, Q, R, u_max,  $\rho$ , eps, max_iters):
60     """Solve the cart-pole swing-up problem via SCP.
61
62     Arguments
63     -----
64     f: callable
65         A function describing the discrete-time dynamics, such that
66         's[k+1] = f(s[k], u[k])'.
67     s0: numpy.ndarray
68         The initial state (1-D).
69     s_goal: numpy.ndarray
70         The goal state (1-D).
71     N: int
72         The time horizon of the LQR cost function.
73     P: numpy.ndarray
74         The terminal state cost matrix (2-D).
75     Q: numpy.ndarray
76         The state stage cost matrix (2-D).
77     R: numpy.ndarray
78         The control stage cost matrix (2-D).
79     u_max: float
80         The bound defining the control set '[-u_max, u_max]'.
81      $\rho$ : float
82         Trust region radius.
83     eps: float
84         Termination threshold for SCP.
85     max_iters: int
86         Maximum number of SCP iterations.
87
88     Returns
89     -----
90     s: numpy.ndarray
91         A 2-D array where 's[k]' is the open-loop state at time step 'k',
92         for 'k=0, 1, ..., N-1'.
93     u: numpy.ndarray
94         A 2-D array where 'u[k]' is the open-loop state at time step 'k',
95         for 'k=0, 1, ..., N-1'.
96     J: numpy.ndarray
97         A 1-D array where 'J[i]' is the SCP sub-problem cost after the i-th
98         iteration, for 'i=0, 1, ..., (iteration when convergence occurred)'.
99     """
100     n = Q.shape[0] # state dimension
101     m = R.shape[0] # control dimension
102
103     # Initialize dynamically feasible nominal trajectories
104     u = np.zeros((N, m))
105     s = np.zeros((N + 1, n))
106     s[0] = s0
107     for k in range(N):
108         s[k + 1] = fd(s[k], u[k])
109
110     # Do SCP until convergence or maximum number of iterations is reached
111     converged = False
112     J = np.zeros(max_iters + 1)

```

```

113 J[0] = np.inf
114 for i in (prog_bar := tqdm(range(max_iters))):
115     s, u, J[i + 1] = scp_iteration(f, s0, s_goal, s, u, N, P, Q, R, u_max,  $\rho$ )
116     dJ = np.abs(J[i + 1] - J[i])
117     prog_bar.set_postfix({"objective_change": "{:.5f}".format(dJ)})
118     if dJ < eps:
119         converged = True
120         print("SCP converged after {} iterations.".format(i))
121         break
122 if not converged:
123     raise RuntimeError("SCP did not converge!")
124 J = J[1 : i + 1]
125 return s, u, J
126
127
128 def scp_iteration(f, s0, s_goal, s_prev, u_prev, N, P, Q, R, u_max,  $\rho$ ):
129     """Solve a single SCP sub-problem for the cart-pole swing-up problem.
130
131     Arguments
132     -----
133     f: callable
134         A function describing the discrete-time dynamics, such that
135         's[k+1] = f(s[k], u[k])'.
136     s0: numpy.ndarray
137         The initial state (1-D).
138     s_goal: numpy.ndarray
139         The goal state (1-D).
140     s_prev: numpy.ndarray
141         The state trajectory around which the problem is convexified (2-D).
142     u_prev: numpy.ndarray
143         The control trajectory around which the problem is convexified (2-D).
144     N: int
145         The time horizon of the LQR cost function.
146     P: numpy.ndarray
147         The terminal state cost matrix (2-D).
148     Q: numpy.ndarray
149         The state stage cost matrix (2-D).
150     R: numpy.ndarray
151         The control stage cost matrix (2-D).
152     u_max: float
153         The bound defining the control set '[-u_max, u_max]'.
154      $\rho$ : float
155         Trust region radius.
156
157     Returns
158     -----
159     s: numpy.ndarray
160         A 2-D array where 's[k]' is the open-loop state at time step 'k',
161         for 'k=0, 1, ..., N-1'.
162     u: numpy.ndarray
163         A 2-D array where 'u[k]' is the open-loop state at time step 'k',
164         for 'k=0, 1, ..., N-1'.
165     J: float
166         The SCP sub-problem cost.
167     """
168     A, B, c = affinize(f, s_prev[:-1], u_prev)
169     A, B, c = np.array(A), np.array(B), np.array(c)
170     n = Q.shape[0]
171     m = R.shape[0]
172     s_cvx = cvx.Variable((N + 1, n))
173     u_cvx = cvx.Variable((N, m))
174
175     # PART (c) #####

```

```

176 # INSTRUCTIONS: Construct the convex SCP sub-problem.
177 objective = cvx.quad_form(s_cvx[N] - s_goal, P)
178 constraints = [s_cvx[0] == s0]
179 for k in range(N):
180     objective += cvx.quad_form(s_cvx[k] - s_goal, Q)
181     objective += cvx.quad_form(u_cvx[k], R)
182     constraints += [
183         s_cvx[k + 1] == A[k] @ s_cvx[k] + B[k] @ u_cvx[k] + c[k],
184         cvx.abs(u_cvx[k, 0]) <= u_max,
185         cvx.norm_inf(s_cvx[k] - s_prev[k]) <= ρ,
186         cvx.norm_inf(u_cvx[k] - u_prev[k]) <= ρ,
187     ]
188 constraints.append(cvx.norm_inf(s_cvx[N] - s_prev[N]) <= ρ)
189 # END PART (c) #####
190
191 prob = cvx.Problem(cvx.Minimize(objective), constraints)
192 prob.solve()
193 if prob.status != "optimal":
194     raise RuntimeError("SCP_solve_failed. Problem status: " + prob.status)
195 s = s_cvx.value
196 u = u_cvx.value
197 J = prob.objective.value
198 return s, u, J
199
200
201 def cartpole(s, u):
202     """Compute the cart-pole state derivative."""
203     mp = 1.0 # pendulum mass
204     mc = 4.0 # cart mass
205     L = 1.0 # pendulum length
206     g = 9.81 # gravitational acceleration
207
208     x, θ, dx, dθ = s
209     sinθ, cosθ = jnp.sin(θ), jnp.cos(θ)
210     h = mc + mp * (sinθ**2)
211     ds = jnp.array(
212         [
213             dx,
214             dθ,
215             (mp * sinθ * (L * (dθ**2) + g * cosθ) + u[0]) / h,
216             -((mc + mp) * g * sinθ + mp * L * (dθ**2) * sinθ * cosθ + u[0] * cosθ)
217             / (h * L),
218         ]
219     )
220     return ds
221
222
223 def discretize(f, dt):
224     """Discretize continuous-time dynamics f via Runge-Kutta integration."""
225
226     def integrator(s, u, dt=dt):
227         k1 = dt * f(s, u)
228         k2 = dt * f(s + k1 / 2, u)
229         k3 = dt * f(s + k2 / 2, u)
230         k4 = dt * f(s + k3, u)
231         return s + (k1 + 2 * k2 + 2 * k3 + k4) / 6
232
233     return integrator
234
235
236 # Define constants
237 n = 4 # state dimension
238 m = 1 # control dimension

```

```

239 s_goal = np.array([0, np.pi, 0, 0]) # desired upright pendulum state
240 s0 = np.array([0, 0, 0, 0]) # initial downright pendulum state
241 dt = 0.1 # discrete time resolution
242 T = 10.0 # total simulation time
243 P = 1e3 * np.eye(n) # terminal state cost matrix
244 Q = np.diag([1e-2, 1.0, 1e-3, 1e-3]) # state cost matrix
245 R = 1e-3 * np.eye(m) # control cost matrix
246 ρ = 1.0 # trust region parameter
247 u_max = 8.0 # control effort bound
248 eps = 5e-1 # convergence tolerance
249 max_iters = 100 # maximum number of SCP iterations
250 animate = True # flag for animation
251
252 # Initialize the discrete-time dynamics
253 fd = jax.jit(discretize(cartpole, dt))
254
255 # Solve the swing-up problem with SCP
256 t = np.arange(0.0, T + dt, dt)
257 N = t.size - 1
258 s, u, J = solve_swingup_scp(fd, s0, s_goal, N, P, Q, R, u_max, ρ, eps, max_iters)
259
260 # Simulate open-loop control
261 for k in range(N):
262     s[k + 1] = fd(s[k], u[k])
263
264 # Plot state and control trajectories
265 fig, ax = plt.subplots(1, n + m, dpi=150, figsize=(15, 2))
266 plt.subplots_adjust(wspace=0.45)
267 labels_s = (r"$x(t)$", r"$\theta(t)$", r"$\dot{x}(t)$", r"$\dot{\theta}(t)$")
268 labels_u = (r"$u(t)$",)
269 for i in range(n):
270     ax[i].plot(t, s[:, i])
271     ax[i].axhline(s_goal[i], linestyle="--", color="tab:orange")
272     ax[i].set_xlabel(r"$t$")
273     ax[i].set_ylabel(labels_s[i])
274 for i in range(m):
275     ax[n + i].plot(t[:-1], u[:, i])
276     ax[n + i].axhline(u_max, linestyle="--", color="tab:orange")
277     ax[n + i].axhline(-u_max, linestyle="--", color="tab:orange")
278     ax[n + i].set_xlabel(r"$t$")
279     ax[n + i].set_ylabel(labels_u[i])
280 plt.savefig(FIG_DIR / "cartpole_swingup_constrained.png", bbox_inches="tight")
281
282 # Plot cost history over SCP iterations
283 fig, ax = plt.subplots(1, 1, dpi=150, figsize=(8, 5))
284 ax.semilogy(J)
285 ax.set_xlabel(r"SCP_iteration_{$i}$")
286 ax.set_ylabel(r"SCP_cost_{$J(\bar{x}^{(i)}, \bar{u}^{(i)})}$")
287 plt.savefig(FIG_DIR / "cartpole_swingup_constrained_cost.png", bbox_inches="tight")
288 # plt.show()
289
290 # Animate the solution
291 if animate:
292     fig, ani = animate_cartpole(t, s[:, 0], s[:, 1])
293     ani.save(FIG_DIR / "cartpole_swingup_constrained.mp4", writer="ffmpeg")
294     # plt.show()

```

cartpole_balance.py

Answers 2.3(a), 2.3(b), 2.3(c), and 2.3(d).

```

1  """
2  Solution code for the problem "Cart-pole balance".
3
4  Autonomous Systems Lab (ASL), Stanford University
5  """
6
7  import numpy as np
8  from scipy.integrate import odeint
9  import matplotlib.pyplot as plt
10 from pathlib import Path
11
12 from animations import animate_cartpole
13
14 # Constants
15 n = 4 # state dimension
16 m = 1 # control dimension
17 mp = 2.0 # pendulum mass
18 mc = 10.0 # cart mass
19 L = 1.0 # pendulum length
20 g = 9.81 # gravitational acceleration
21 dt = 0.1 # discretization time step
22 animate = True # whether or not to animate results
23
24 FIG_DIR = Path(__file__).resolve().parents[1] / "latex" / "figures"
25 FIG_DIR.mkdir(parents=True, exist_ok=True)
26
27 def cartpole(s: np.ndarray, u: np.ndarray) -> np.ndarray:
28     """Compute the cart-pole state derivative
29
30     Args:
31         s (np.ndarray): The cartpole state: [x, theta, x_dot, theta_dot], shape (n,)
32         u (np.ndarray): The cartpole control: [F_x], shape (m,)
33
34     Returns:
35         np.ndarray: The state derivative, shape (n,)
36     """
37     x, theta, dx, dtheta = s
38     sintheta, costheta = np.sin(theta), np.cos(theta)
39     h = mc + mp * (sintheta**2)
40     ds = np.array(
41         [
42             dx,
43             dtheta,
44             (mp * sintheta * (L * (dtheta**2) + g * costheta) + u[0]) / h,
45             -((mc + mp) * g * sintheta + mp * L * (dtheta**2) * sintheta * costheta + u[0] * costheta)
46             / (h * L),
47         ]
48     )
49     return ds
50
51
52 def reference(t: float) -> np.ndarray:
53     """Compute the reference state (s_bar) at time t
54
55     Args:
56         t (float): Evaluation time
57
58     Returns:
59         np.ndarray: Reference state, shape (n,)
60     """
61     a = 10.0 # Amplitude
62     T = 10.0 # Period

```

```

63
64 # PART (d) #####
65 # INSTRUCTIONS: Compute the reference state for a given time
66 omega = 2 * np.pi / T
67 return np.array([a * np.sin(omega * t), np.pi, a * omega * np.cos(omega * t), 0.0])
68 # END PART (d) #####
69
70
71 def ricatti_recursion(
72     A: np.ndarray, B: np.ndarray, Q: np.ndarray, R: np.ndarray
73 ) -> np.ndarray:
74     """Compute the gain matrix K through Ricatti recursion
75
76     Args:
77         A(np.ndarray): Dynamics matrix, shape (n, n)
78         B(np.ndarray): Controls matrix, shape (n, m)
79         Q(np.ndarray): State cost matrix, shape (n, n)
80         R(np.ndarray): Control cost matrix, shape (m, m)
81
82     Returns:
83         np.ndarray: Gain matrix K, shape (m, n)
84     """
85     eps = 1e-4 # Riccati recursion convergence tolerance
86     max_iters = 1000 # Riccati recursion maximum number of iterations
87     P_prev = np.zeros((n, n)) # initialization
88     converged = False
89     for i in range(max_iters):
90         # PART (b) #####
91         # INSTRUCTIONS: Apply the Riccati equation until convergence
92         K = -np.linalg.solve(R + B.T @ P_prev @ B, B.T @ P_prev @ A)
93         P = Q + A.T @ P_prev @ (A + B @ K)
94         if np.max(np.abs(P - P_prev)) < eps:
95             converged = True
96             break
97         P_prev = P
98         # END PART (b) #####
99     if not converged:
100         raise RuntimeError("Ricatti recursion did not converge!")
101     print("K:", K)
102     return K
103
104
105 def simulate(
106     t: np.ndarray, s_ref: np.ndarray, u_ref: np.ndarray, s0: np.ndarray, K: np.ndarray
107 ) -> tuple[np.ndarray, np.ndarray]:
108     """Simulate the cartpole
109
110     Args:
111         t(np.ndarray): Evaluation times, shape (num_timesteps,)
112         s_ref(np.ndarray): Reference state s_bar, evaluated at each time t. Shape (
113             num_timesteps, n)
114         u_ref(np.ndarray): Reference control u_bar, shape (m,)
115         s0(np.ndarray): Initial state, shape (n,)
116         K(np.ndarray): Feedback gain matrix (Ricatti recursion result), shape (m, n)
117
118     Returns:
119         tuple[np.ndarray, np.ndarray]: Tuple of:
120         np.ndarray: The state history, shape (num_timesteps, n)
121         np.ndarray: The control history, shape (num_timesteps, m)
122     """
123
124     def cartpole_wrapper(s, t, u):
125         """Helper function to get cartpole() into a form preferred by odeint, which

```

```

125         expects_t_as_the_second_arg"""
126         return cartpole(s, u)
127
128     # PART (c) #####
129     # INSTRUCTIONS: Complete the function to simulate the cartpole system
130     # Hint: use the cartpole wrapper above with odeint
131     s = np.zeros((t.size, n))
132     u = np.zeros((t.size, m))
133     s[0] = s0
134     for k in range(t.size - 1):
135         u[k] = u_ref + K @ (s[k] - s_ref[k])
136         s[k + 1] = odeint(cartpole_wrapper, s[k], t[k : k + 2], args=(u[k],))[1]
137     u[-1] = u[-2]
138     # END PART (c) #####
139     return s, u
140
141 def compute_lti_matrices() -> tuple[np.ndarray, np.ndarray]:
142     """Compute the linearized dynamics matrices A and B of the LTI system
143
144     Returns:
145     tuple[np.ndarray, np.ndarray]: Tuple of:
146     np.ndarray: The A (dynamics) matrix, shape (n, n)
147     np.ndarray: The B (controls) matrix, shape (n, m)
148     """
149     # PART (a) #####
150     # INSTRUCTIONS: Construct the A and B matrices
151     a = mp * g / mc
152     b = (mc + mp) * g / (mc * L)
153     F = np.array(
154         [
155             [0.0, 0.0, 1.0, 0.0],
156             [0.0, 0.0, 0.0, 1.0],
157             [0.0, a, 0.0, 0.0],
158             [0.0, b, 0.0, 0.0],
159         ]
160     )
161     A = np.eye(n) + dt * F
162     B = dt * np.array([[0.0], [0.0], [1.0 / mc], [1.0 / (mc * L)]])
163     # END PART (a) #####
164     return A, B
165
166
167 def plot_state_and_control_history(
168     s: np.ndarray, u: np.ndarray, t: np.ndarray, s_ref: np.ndarray, name: str
169 ) -> None:
170     """Helper function for cartpole visualization
171
172     Args:
173     s(np.ndarray): State history, shape (num_timesteps, n)
174     u(np.ndarray): Control history, shape (num_timesteps, m)
175     t(np.ndarray): Times, shape (num_timesteps,)
176     s_ref(np.ndarray): Reference state s_bar, evaluated at each time t. Shape (
177         num_timesteps, n)
178     name(str): Filename prefix for saving figures
179     """
180     fig, axes = plt.subplots(1, n + m, dpi=150, figsize=(15, 2))
181     plt.subplots_adjust(wspace=0.35)
182     labels_s = (r"$x(t)$", r"$\theta(t)$", r"$\dot{x}(t)$", r"$\dot{\theta}(t)$")
183     labels_u = (r"$u(t)$",)
184     for i in range(n):
185         axes[i].plot(t, s[:, i])
186         axes[i].plot(t, s_ref[:, i], "--")

```



```

186         axes[i].set_xlabel(r"$t$")
187         axes[i].set_ylabel(labels_s[i])
188     for i in range(m):
189         axes[n + i].plot(t, u[:, i])
190         axes[n + i].set_xlabel(r"$t$")
191         axes[n + i].set_ylabel(labels_u[i])
192     plt.savefig(FIG_DIR / f"{name}.png", bbox_inches="tight")
193     # plt.show()
194
195     if animate:
196         fig, ani = animate_cartpole(t, s[:, 0], s[:, 1])
197         ani.save(FIG_DIR / f"{name}.mp4", writer="ffmpeg")
198         # plt.show()
199
200
201 def main():
202     # Part A
203     A, B = compute_lti_matrices()
204
205     # Part B
206     Q = np.eye(n) # state cost matrix
207     R = np.eye(m) # control cost matrix
208     K = ricatti_recursion(A, B, Q, R)
209
210     # Part C
211     t = np.arange(0.0, 30.0, 1 / 10)
212     s_ref = np.array([0.0, np.pi, 0.0, 0.0]) * np.ones((t.size, 1))
213     u_ref = np.array([0.0])
214     s0 = np.array([0.0, 3 * np.pi / 4, 0.0, 0.0])
215     s, u = simulate(t, s_ref, u_ref, s0, K)
216     plot_state_and_control_history(s, u, t, s_ref, "cartpole_balance")
217
218     # Part D
219     # Note: t, u_ref unchanged from part c
220     s_ref = np.array([reference(ti) for ti in t])
221     s0 = np.array([0.0, np.pi, 0.0, 0.0])
222     s, u = simulate(t, s_ref, u_ref, s0, K)
223     plot_state_and_control_history(s, u, t, s_ref, "cartpole_balance_tv")
224
225
226 if __name__ == "__main__":
227     main()

```

cartpole_swingup.py

Answers 2.4(a), 2.4(c), and 2.4(d).

```

1  """
2  Starter code for the problem "Cart-pole swing-up".
3
4  Autonomous Systems Lab (ASL), Stanford University
5  """
6
7  import time
8  from pathlib import Path
9
10 from animations import animate_cartpole
11
12 import jax
13 import jax.numpy as jnp
14

```

```

15 import matplotlib.pyplot as plt
16
17 import numpy as np
18
19 from scipy.integrate import odeint
20
21 FIG_DIR = Path(__file__).resolve().parents[1] / "latex" / "figures"
22 FIG_DIR.mkdir(parents=True, exist_ok=True)
23
24 def linearize(f, s, u):
25     """Linearize the function f(s,u) around (s,u).
26
27     Arguments
28     -----
29     f: callable
30         A nonlinear function with call signature f(s,u).
31     s: numpy.ndarray
32         The state (1-D).
33     u: numpy.ndarray
34         The control input (1-D).
35
36     Returns
37     -----
38     A: numpy.ndarray
39         The Jacobian of f at (s,u), with respect to s.
40     B: numpy.ndarray
41         The Jacobian of f at (s,u), with respect to u.
42     """
43     # WRITE YOUR CODE BELOW #####
44     # INSTRUCTIONS: Use JAX to compute 'A' and 'B' in one line.
45     A, B = jax.jacobian(f, argnums=(0, 1))(s, u)
46     #####
47     return A, B
48
49
50 def ilqr(f, s0, s_goal, N, Q, R, QN, eps=1e-3, max_iters=1000):
51     """Compute the iLQR set-point tracking solution.
52
53     Arguments
54     -----
55     f: callable
56         A function describing the discrete-time dynamics, such that
57         s[k+1] = f(s[k], u[k]).
58     s0: numpy.ndarray
59         The initial state (1-D).
60     s_goal: numpy.ndarray
61         The goal state (1-D).
62     N: int
63         The time horizon of the LQR cost function.
64     Q: numpy.ndarray
65         The state cost matrix (2-D).
66     R: numpy.ndarray
67         The control cost matrix (2-D).
68     QN: numpy.ndarray
69         The terminal state cost matrix (2-D).
70     eps: float, optional
71         Termination threshold for iLQR.
72     max_iters: int, optional
73         Maximum number of iLQR iterations.
74
75     Returns
76     -----
77     s_bar: numpy.ndarray

```

```

78     A_2 = D_array where s_bar[k] 'is the nominal state at time step k',
79     for k=0,1,...,N-1'
80     u_bar = numpy.ndarray
81     A_2 = D_array where u_bar[k] 'is the nominal control at time step k',
82     for k=0,1,...,N-1'
83     Y = numpy.ndarray
84     A_3 = D_array where Y[k] 'is the matrix gain term of the iLQR control
85     law at time step k', for k=0,1,...,N-1'
86     y = numpy.ndarray
87     A_2 = D_array where y[k] 'is the offset term of the iLQR control law
88     at time step k', for k=0,1,...,N-1'
89     """
90     if max_iters <= 1:
91         raise ValueError("Argument 'max_iters' must be at least 1.")
92     n = Q.shape[0] # state dimension
93     m = R.shape[0] # control dimension
94
95     # Initialize gains 'Y' and offsets 'y' for the policy
96     Y = np.zeros((N, m, n))
97     y = np.zeros((N, m))
98
99     # Initialize the nominal trajectory '(s_bar, u_bar)', and the
100    # deviations '(ds, du)'
101    u_bar = np.zeros((N, m))
102    s_bar = np.zeros((N + 1, n))
103    s_bar[0] = s0
104    for k in range(N):
105        s_bar[k + 1] = f(s_bar[k], u_bar[k])
106    ds = np.zeros((N + 1, n))
107    du = np.zeros((N, m))
108
109    def cost(s, u):
110        J = 0.5 * (s[N] - s_goal) @ QN @ (s[N] - s_goal)
111        for k in range(N):
112            J += 0.5 * (s[k] - s_goal) @ Q @ (s[k] - s_goal)
113            J += 0.5 * u[k] @ R @ u[k]
114        return J
115
116    # iLQR loop
117    converged = False
118    for _ in range(max_iters):
119        # Linearize the dynamics at each step 'k' of '(s_bar, u_bar)'
120        A, B = jax.vmap(linearize, in_axes=(None, 0, 0))(f, s_bar[:-1], u_bar)
121        A, B = np.array(A), np.array(B)
122
123        # PART (c) #####
124        # INSTRUCTIONS: Update 'Y', 'y', 'ds', 'du', 's_bar', and 'u_bar'.
125        q = [Q @ (s_bar[k] - s_goal) for k in range(N)]
126        r = [R @ u_bar[k] for k in range(N)]
127        P = QN.copy()
128        p = QN @ (s_bar[N] - s_goal)
129
130        for k in reversed(range(N)):
131            G = R + B[k].T @ P @ B[k]
132            Y[k] = -np.linalg.solve(G, B[k].T @ P @ A[k])
133            y[k] = -np.linalg.solve(G, r[k] + B[k].T @ p)
134            p = q[k] + A[k].T @ p + A[k].T @ P @ B[k] @ y[k]
135            P = Q + A[k].T @ P @ (A[k] + B[k] @ Y[k])
136
137        ds.fill(0.0)
138        du.fill(0.0)
139        for k in range(N):
140            du[k] = Y[k] @ ds[k] + y[k]

```

```

141         ds[k + 1] = A[k] @ ds[k] + B[k] @ du[k]
142
143     J_prev = cost(s_bar, u_bar)
144     accepted = False
145     for  $\alpha$  in (1.0, 0.5, 0.25, 0.1, 0.05, 0.01):
146         u_next = u_bar +  $\alpha$  * du
147         s_next = np.zeros_like(s_bar)
148         s_next[0] = s0
149         for k in range(N):
150             s_next[k + 1] = f(s_next[k], u_next[k])
151             if np.all(np.isfinite(s_next)) and cost(s_next, u_next) < J_prev:
152                 accepted = True
153                 break
154         if not accepted:
155              $\alpha$  = 0.0
156             u_next = u_bar
157             s_next = s_bar
158             du.fill(0.0)
159     u_bar = u_next
160     s_bar = s_next
161     #####
162
163     if np.max(np.abs( $\alpha$  * du)) < eps:
164         converged = True
165         break
166     if not converged:
167         raise RuntimeError("iLQR did not converge!")
168     return s_bar, u_bar, Y, y
169
170
171 def cartpole(s, u):
172     """Compute the cart-pole state derivative."""
173     mp = 2.0 # pendulum mass
174     mc = 10.0 # cart mass
175     L = 1.0 # pendulum length
176     g = 9.81 # gravitational acceleration
177
178     x,  $\theta$ , dx, d $\theta$  = s
179     sin $\theta$ , cos $\theta$  = jnp.sin( $\theta$ ), jnp.cos( $\theta$ )
180     h = mc + mp * (sin $\theta$ **2)
181     ds = jnp.array(
182         [
183             dx,
184             d $\theta$ ,
185             (mp * sin $\theta$  * (L * (d $\theta$ **2) + g * cos $\theta$ ) + u[0]) / h,
186             -((mc + mp) * g * sin $\theta$  + mp * L * (d $\theta$ **2) * sin $\theta$  * cos $\theta$  + u[0] * cos $\theta$ )
187             / (h * L),
188         ]
189     )
190     return ds
191
192
193 # Define constants
194 n = 4 # state dimension
195 m = 1 # control dimension
196 Q = np.diag(np.array([10.0, 10.0, 2.0, 2.0])) # state cost matrix
197 R = 1e-2 * np.eye(m) # control cost matrix
198 QN = 1e2 * np.eye(n) # terminal state cost matrix
199 s0 = np.array([0.0, 0.0, 0.0, 0.0]) # initial state
200 s_goal = np.array([0.0, np.pi, 0.0, 0.0]) # goal state
201 T = 10.0 # simulation time
202 dt = 0.1 # sampling time
203 animate = False # flag for animation

```

```

204 closed_loop = False # flag for closed-loop control
205
206 # Initialize continuous-time and discretized dynamics
207 f = jax.jit(cartpole)
208 fd = jax.jit(lambda s, u, dt=dt: s + dt * f(s, u))
209
210 # Compute the iLQR solution with the discretized dynamics
211 print("Computing iLQR solution...", end="", flush=True)
212 start = time.time()
213 t = np.arange(0.0, T, dt)
214 N = t.size - 1
215 s_bar, u_bar, Y, y = ilqr(fd, s0, s_goal, N, Q, R, QN)
216 print("done!({:.2f}s)".format(time.time() - start), flush=True)
217
218 def simulate_and_plot(closed_loop):
219     """Simulate either the open-loop sequence or closed-loop iLQR policy."""
220     print("Simulating {}...".format("closed-loop" if closed_loop else "open-loop"), end=
221           "", flush=True)
222     start = time.time()
223     s = np.zeros((N + 1, n))
224     u = np.zeros((N, m))
225     s[0] = s0
226     for k in range(N):
227         # PART (d) #####
228         # INSTRUCTIONS: Compute either the closed-loop or open-loop value of
229         # 'u[k]', depending on the Boolean flag 'closed_loop'.
230         if closed_loop:
231             u[k] = u_bar[k] + Y[k] @ (s[k] - s_bar[k]) + y[k]
232         else: # do open-loop control
233             u[k] = u_bar[k]
234             #####
235             s[k + 1] = odeint(lambda s, t: f(s, u[k]), s[k], t[k : k + 2])[1]
236     print("done!({:.2f}s)".format(time.time() - start), flush=True)
237
238     fig, axes = plt.subplots(1, n + m, dpi=150, figsize=(15, 2))
239     plt.subplots_adjust(wspace=0.45)
240     labels_s = (r"$x(t)$", r"$\theta(t)$", r"$\dot{x}(t)$", r"$\dot{\theta}(t)$")
241     labels_u = (r"$u(t)$",)
242     for i in range(n):
243         axes[i].plot(t, s[:, i])
244         axes[i].set_xlabel(r"$t$")
245         axes[i].set_ylabel(labels_s[i])
246     for i in range(m):
247         axes[n + i].plot(t[:-1], u[:, i])
248         axes[n + i].set_xlabel(r"$t$")
249         axes[n + i].set_ylabel(labels_u[i])
250     if closed_loop:
251         plt.savefig(FIG_DIR / "cartpole_swingup_cl.png", bbox_inches="tight")
252     else:
253         plt.savefig(FIG_DIR / "cartpole_swingup_ol.png", bbox_inches="tight")
254     plt.show()
255
256     if animate:
257         fig, ani = animate_cartpole(t, s[:, 0], s[:, 1])
258         ani.save(FIG_DIR / "cartpole_swingup.mp4", writer="ffmpeg")
259         plt.show()
260
261 simulate_and_plot(False)
262 simulate_and_plot(True)

```